Huawei Access Node WFD-ACC-18 Routing Protocol Flap Resolution SOP

Node: WFD-ACC-18 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_9621

Description

The Huawei Access Node WFD-ACC-18 at Whitefield POP is experiencing a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) failure, indicated by Alarm Code ALARM_9621 with a Severity of Warning. This failure domain occurs due to the inherent dynamic nature of routing protocols, which can lead to instability and churn in the control plane. Specifically, on Huawei Access Nodes, BGP route churn can be triggered by updates to the routing table, OSPF adjacency loss can be caused by interface flapping or neighbor mismatch, and ISIS LSP flooding can result from misconfigured or malfunctioning interfaces. The root cause of this failure must be identified and addressed to prevent further instability and potential outages.

Pre-requisite Checks

- Verify that the Whitefield POP is accessible and that remote hands are available for assistance.

- Check the status of redundant links to ensure that they are operational and can take over in case of a failure.

- Review the system logs for any errors or anomalies related to the routing protocols.

Potential Root Causes

Cause	Probability
BGP Route Churn	High
OSPF Adjacency Loss	Medium
ISIS LSP Flooding	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp routing-table, Expected Output: showing the current BGP routing table with updates and churn, Description: This command displays the current BGP routing table, including any updates or churn, which can help identify the root cause of the failure.

Command: display ospf routing-table, Expected Output: showing the current OSPF routing table with adjacency loss, Description: This command displays the current OSPF routing table, including any adjacency loss, which can help identify the root cause of the failure.

Command: display isis database, Expected Output: showing the current ISIS database with LSP flooding, Description: This command displays the current ISIS database, including any LSP flooding, which can help identify the root cause of the failure.
```

Resolution Steps

Step 1: Step: Verify BGP Route Churn, Command: display bgp routing-table | include churn, Description: Verify that there are no updates or churn in the BGP routing table. If there are, troubleshoot and resolve the issue before proceeding.

Step 2: Step: Verify OSPF Adjacency Loss, Command: display ospf routing-table | include adjacency-loss, Description: Verify that there are no adjacency loss issues in the OSPF routing table. If there are, troubleshoot and resolve the issue before proceeding.

Step 3: Step: Verify ISIS LSP Flooding, Command: display isis database | include lsp-flooding, Description: Verify that there are no LSP flooding issues in the ISIS database. If there are, troubleshoot and resolve the issue before proceeding.

Step 4: Step: Clear BGP Route Churn, Command: clear bgp routing-table, Description: Clear any BGP route churn issues by restarting the BGP process.

Step 5: Step: Clear OSPF Adjacency Loss, Command: clear ospf adjacency-loss, Description: Clear any OSPF adjacency loss issues by restarting the OSPF process.

Step 6: Step: Clear ISIS LSP Flooding, Command: clear isis lsp-flooding, Description: Clear any ISIS LSP flooding issues by restarting the ISIS process.

Post-Check Verification

```
Command: display bgp routing-table | include churn, Expected Output:
no updates or churn in the BGP routing table, Description: Verify that
there are no updates or churn in the BGP routing table after resolving
the issue.

Command: display ospf routing-table | include adjacency-loss, Expected
Output: no adjacency loss issues in the OSPF routing table,
Description: Verify that there are no adjacency loss issues in the
OSPF routing table after resolving the issue.

Command: display isis database | include lsp-flooding, Expected
Output: no LSP flooding issues in the ISIS database, Description:
Verify that there are no LSP flooding issues in the ISIS database
after resolving the issue.
```

Roll-back Procedures

- Step: Restore BGP Routing Table, Command: undo clear bgp routing-table, Description: Restore the BGP routing table to its previous state if the fix fails.

- Step: Restore OSPF Adjacency Loss, Command: undo clear ospf adjacency-loss, Description: Restore the OSPF adjacency loss issue to its previous state if the fix fails.

- Step: Restore ISIS LSP Flooding, Command: undo clear isis lsp-flooding, Description: Restore the ISIS LSP flooding issue to its previous state if the fix fails.